

Utility Usage Prediction Tool – Project 1

AI/ML Internship Project Report

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Utility usage prediction is an important application of Machine Learning that helps estimate future consumption based on historical records.

This project predicts utility usage using previous monthly data and a Linear Regression model.

Objectives

- Learn Python programming.
- Learn CSV file handling.
- Train a Machine Learning model.
- Predict utility usage.
- Handle user input and exceptions.

Tools & Technologies

Programming Language

- Python

Libraries

- Pandas
- Scikit-learn

Platform

- Google Colab

Algorithm

- Linear Regression

Dataset

The dataset contains two columns:

Month	Usage
1	120
2	135
3	150
4	160
5	180
6	200
7	210
8	205
9	195

10	180
11	170
12	160

The data is stored in a CSV file named **utility_usage.csv**.

The project follows these steps:

1. Read CSV file.
2. Train Linear Regression Model.
3. Display Menu.
4. Add Data.
5. Update Data.
6. Predict Usage.
7. Save Updated Data.

Features

- Menu Driven Application
- CSV File Handling
- Add Records
- Update Records
- Machine Learning Prediction
- Exception Handling

The project uses **Linear Regression**.

Input

- Month

Output

- Predicted Utility Usage

The model learns from historical utility consumption and predicts future usage.

Result

The project successfully predicts utility usage based on user input.

Sample Output:

Enter Month: 9

Predicted Utility Usage: 189.75

Future Scope

- Graphical User Interface
- Better Prediction Algorithms
- Cloud Database Integration
- Multiple Utility Prediction

The Utility Usage Prediction Tool demonstrates the implementation of Machine Learning using Linear Regression. It provides a simple and effective way to predict utility consumption and helps understand the basic workflow of AI/ML applications.